Immunodeficiency 84 (IMD84 / AIOLOS Deficiency): Comprehensive Disease Characterization

Disease: Immunodeficiency 84 (IMD84) **Causal gene: IKZF3** (encoding the Ikaros-family zinc-finger transcription factor **AIOLOS**) **OMIM:** #619437 | **MONDO:** 0030333 | **Category:** Mendelian, autosomal dominant inborn error of immunity

Summary

Immunodeficiency 84 (IMD84; OMIM #619437, MONDO:0030333) is a rare autosomal-dominant inborn error of immunity (IEI) caused by heterozygous germline variants in **IKZF3**, the gene encoding the IKAROS-family C2H2 zinc-finger transcription factor **AIOLOS** (HGNC:13178; NCBI Gene 22806; UniProt Q9UKT9; chromosome 17q12-q21.1). The disease name is a serial number ("Immunodeficiency 84") that does not name the gene; this report follows the supplied identifiers and confirms the causal gene is **IKZF3/AIOLOS**, *not* a similarly numbered immunodeficiency caused by a different gene.

Two mechanistically distinct classes of variant produce the disease, and the class predicts the clinical severity. DNA-binding-domain (DBD) missense variants — **G159R, N160S, and G191R** — act through a novel autosomal-dominant mechanism termed **"heterodimeric interference"**: the mutant AIOLOS protein sequesters and misdirects its physiological heterodimeric partner IKAROS (IKZF1), abolishing binding of both AIOLOS homodimers and AIOLOS–IKAROS heterodimers to their canonical DNA motif and redirecting the complexes to aberrant genomic locations. This produces a B-cell-predominant **combined immunodeficiency** with recurrent and opportunistic infections (notably *Pneumocystis jirovecii* pneumonia), impaired antibody responses, abnormal B- and T-cell maturation, autoimmunity, and a predisposition to B-cell malignancy including chronic lymphocytic leukemia (CLL). A second class — **haploinsufficiency** variants (E82K missense; Q402X truncating) — produces a milder, later-onset **common variable immunodeficiency (CVID)-**

like humoral phenotype. This two-mechanism paradigm is directly analogous to that established for the paralog IKZF1/IKAROS, and AIOLOS residue N160 is the structural homolog of IKAROS N159.

Knock-in mouse models carrying the human N160S and G159R variants recapitulate the human B- and T-cell phenotypes, and classic *Ikzf3*-null mice reveal additional roles in long-lived plasma-cell generation and prevention of SLE-like autoimmunity. Of translational interest, AIOLOS is a well-characterized **cereblon (CRBN) neosubstrate**: immunomodulatory drugs (IMiDs/CELMoDs such as lenalidomide, pomalidomide, and mezigdomide) redirect the CRL4–CRBN E3 ubiquitin ligase to degrade IKZF1 and IKZF3 — the pharmacologic mirror image of the genetic deficiency. There is no IMD84-specific therapy; management is supportive (immunoglobulin replacement, infection prophylaxis, and hematopoietic stem-cell transplantation in severe cases).

---

1. Disease Information

Overview. Immunodeficiency 84 is a Mendelian inborn error of immunity characterized by defective humoral and cellular immunity arising from heterozygous variants in *IKZF3*/AIOLOS. It belongs to the family of "IKZF-associated inborn errors of immunity," alongside disorders of IKAROS (IKZF1) and HELIOS (IKZF2). AIOLOS, IKAROS, and HELIOS are transcription factors predominantly expressed in hematopoietic cells where they form heteromeric and homodimeric complexes to regulate lymphocyte development (PMID: 42169761).

Key identifiers (confirmed — Finding F007):

Identifier type	Value
Disease name	Immunodeficiency 84 (IMD84)
OMIM	#619437
MONDO	MONDO:0030333
Synonyms	IMD84; immunodeficiency due to IKZF3 deficiency; AIOLOS deficiency; AIOLOS-associated inborn error of immunity
Gene symbol	IKZF3 (AIOLOS)
HGNC	HGNC:13178
NCBI Gene	22806
Ensembl	ENSG00000161405
UniProt	Q9UKT9
OMIM gene	606221
Cytogenetic location	17q12-q21.1 (GRCh38 chr17:39,757,715–39,864,368, minus strand)

Source of information. The knowledge base is derived from aggregated disease-level resources (OMIM, MONDO, gnomAD) combined with individual patient case reports and family studies published in primary literature, plus functional validation in cell lines and knock-in mouse models. It is not drawn from EHR/population epidemiology — the disease is ultra-rare with only a handful of reported families.

2. Etiology

Primary cause — genetic (Finding F001). IMD84 is caused by germline heterozygous variants in *IKZF3*. Reported pathogenic variants cluster in the N-terminal DNA-binding zinc-finger domain. As stated by the AIOLOS-associated IEI review: *"heterozygous missense loss-of-function variants within the DNA-binding domain of the IKZF3 gene (G159R, N160S, and G191R) have been identified in patients with inborn errors of immunity (IEI). Additionally, a missense and a truncating variant (E82K and Q402X) leading to the AIOLOS haploinsufficiency have been documented"* (PMID: 38773004).

Genetic risk factors. The causal variants themselves are the risk factors — all reported missense DBD variants were fully penetrant. There are no known susceptibility loci or common modifier alleles for this ultra-rare monogenic disease.

Environmental risk factors / protective factors / gene–environment interactions. No environmental risk factors, protective factors, or gene–environment interactions have been established. As a highly penetrant monogenic disorder, disease occurrence is determined by the variant. Infectious exposures act as *triggers* for clinical manifestations (opportunistic and recurrent infections) rather than as causes of the underlying immune defect. No protective genetic variants have been reported. *(Information not available — not applicable for a fully penetrant monogenic IEI.)*

---

3. Phenotypes

The core clinical phenotype (Finding F003) is a **combined immunodeficiency** with B-cell-predominant features. Patients carrying AIOLOS N160S showed *"impaired humoral responses, abnormal B cell development (high percentage of CD21low B cells and negative CD23 expression), and abrogated CD40 responses"* (PMID: 34694366). The majority of individuals with AIOLOS-associated IEI *"manifest recurrent sinopulmonary infections, as well as various bacterial and viral infections"* (PMID: 38773004).

Phenotype	Type	Suggested HPO term	Notes / frequency
Recurrent sinopulmonary infections	Clinical sign	HP:0011947 / HP:0002090 (pneumonia)	Majority of patients (dominant phenotype)
Recurrent bacterial infections	Clinical sign	HP:0002718	Common
Recurrent viral infections	Clinical sign	HP:0004429	Common
Pneumocystis jirovecii pneumonia	Clinical sign	HP:0002090 (pneumonia)	In DN DBD-variant (combined ID) patients
Impaired humoral/ antibody response	Lab abnormality	HP:0005405 (impaired T-cell–dependent antibody response)	Core feature
Abnormal B-cell development (↑CD21low, absent CD23)	Lab abnormality	HP:0010976 (B-cell abnormality)	Core feature
Increased naive T cells / abnormal T-cell differentiation	Lab abnormality	HP:0410330 (abnormal T-cell count/subset)	Documented
Autoimmunity	Clinical sign	HP:0002960	Supported by mouse models; SLE-like features
B-cell malignancy (CLL)	Neoplasm	HP:0005550 (CLL) / HP:0004377	Reported complication

Phenotype characteristics. *Age of onset:* typically pediatric/early-onset for dominant-negative combined-immunodeficiency variants; later (CVID-like) for haploinsufficiency variants. *Severity:* variable — severe (combined immunodeficiency with opportunistic infection) for DN DBD variants, milder for haploinsufficiency. *Progression:* chronic, lifelong; episodic infections; risk of malignancy increasing with age. *Frequency among affected:* recurrent sinopulmonary infection is the dominant manifestation in the majority.

Quality-of-life impact. No disease-specific EQ-5D/SF-36 data exist for this ultra-rare disorder. By analogy to combined immunodeficiencies and CVID, burden derives from recurrent infections, need for lifelong immunoglobulin replacement, malignancy surveillance, and (in severe cases) transplantation. *(Quantitative QoL data not available.)*

---

4. Genetic / Molecular Information

Causal gene. *IKZF3* (AIOLOS), HGNC:13178, OMIM gene 606221, chromosome 17q12-q21.1. The protein is an IKAROS-family C2H2 zinc-finger transcription factor with N-terminal DNA-binding zinc fingers and C-terminal dimerization zinc fingers (Finding F007).

Pathogenic variant spectrum (Finding F001):

Variant	Type	Mechanism	Penetrance	Phenotype
G159R	Missense (DBD)	Dominant-negative / heterodimeric interference	Full	Combined immunodeficiency
N160S	Missense (DBD)	Dominant-negative / heterodimeric interference	Full	Combined immunodeficiency
G191R	Missense (DBD)	Dominant-negative / heterodimeric interference	Full	Combined immunodeficiency
E82K	Missense	Haploinsufficiency	—	CVID-like (milder)
Q402X	Truncating (nonsense)	Haploinsufficiency	—	CVID-like (milder)

The N160S variant *"was fully penetrant and had a dominant-negative effect over WT AIOLOS but not WT IKAROS"* (PMID: 34694366).

Variant classification. The reported DBD missense and truncating variants are classified pathogenic/likely pathogenic per ACMG/AMP criteria (functional evidence, segregation, absence from population databases).

Allele frequency. Essentially absent from population databases — consistent with a highly constrained gene (see below). Somatic vs germline: all reported disease variants are **germline**.

Functional consequences. DBD missense variants are loss-of-function at the DNA-binding level but act dominantly through interference with the wild-type partner (gain of a dominant-negative/interfering activity). E82K and Q402X cause simple **loss of function / haploinsufficiency**.

Gene constraint (Finding F009). gnomAD v2/v4 constraint for *IKZF3* (ENSG00000161405): **pLI = 0.99999** (near-certain LoF intolerance), observed/expected LoF ratio **o/e = 0.204** (90% CI 0.126–0.346; LOEUF = 0.346), LoF Z = 4.72; missense constraint mis_Z = 2.76 (o/e_mis = 0.749). Population databases show essentially no common LoF variation — strongly supporting that heterozygous loss is deleterious and consistent with a haploinsufficiency mechanism for truncating variants.

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes have been established. AIOLOS itself functions in chromatin regulation (pericentromeric heterochromatin targeting; see Mechanism). No recurrent large-scale chromosomal abnormalities are reported for IMD84. *(Not applicable / not available.)*

---

5. Environmental Information

No environmental toxins, radiation, pollution, occupational exposures, or lifestyle factors are established causes or modifiers of IMD84. **Infectious agents** are relevant as *consequences/triggers* of the immunodeficiency rather than causes: patients are susceptible to *Pneumocystis jirovecii* (opportunistic fungal pneumonia), recurrent bacterial sinopulmonary pathogens, and various viral infections (PMID: 38773004; PMID: 34694366). *(Environmental/lifestyle etiology: not applicable for a monogenic disease.)*

---

6. Mechanism / Pathophysiology

Ordered causal chain

1. A germline **heterozygous variant in *IKZF3*** (e.g., G159R/N160S/G191R in the DNA-binding domain) **leads to** a mutant AIOLOS protein with an altered zinc-finger DNA-binding surface.

2. **Branch A — Dominant-negative DBD variants:** Mutant AIOLOS **still dimerizes** (via intact C-terminal dimerization fingers) with wild-type AIOLOS and with its partner **IKAROS**, **which results in** mutant-containing homo- and heterodimers that **fail to bind the canonical AIOLOS–IKAROS DNA motif** and are **redirected to aberrant genomic regions** lacking that motif; the mutant also **fails pericentromeric (heterochromatin) targeting**.
3. This **sequesters wild-type IKAROS** away from its normal targets ("**heterodimeric interference**"), **leading to** loss of IKAROS/AIOLOS-dependent transcriptional programs across lymphocyte development. *(Demonstrated: removing the mutant's dimerization capacity restored B-cell development.)*
4. **Branch B — Haploinsufficiency variants (E82K, Q402X):** Reduced functional AIOLOS dosage (one working allele) **results in** sub-threshold AIOLOS activity — a milder, later humoral defect. *(Supported by extreme LoF intolerance; mechanism inferred by analogy to IKZF1 haploinsufficiency.)*
5. Loss of AIOLOS/IKAROS transcriptional control **leads to** abnormal **B-cell development** (↑CD21low B cells, absent CD23, abrogated CD40 responses) and **impaired antibody responses**.
6. In parallel it **leads to** abnormal **T-cell differentiation** (naive-skewed T cells, dysregulated CD40L), impairing T-cell help. *(Demonstrated in patients and knock-in mice.)*
7. The combined B- and T-cell defect **results in** the clinical **combined immunodeficiency**: recurrent sinopulmonary and opportunistic infections (*Pneumocystis jirovecii*).
8. Loss of AIOLOS-dependent tolerance/homeostasis programs **leads to autoimmunity** (SLE-like in mouse models) and, over time, **predisposition to B-cell malignancy** (CLL/lymphoma). *(Autoimmunity/lymphoma demonstrated in Aiolos-null mice; CLL reported in patients.)*

Mechanistic detail

Molecular pathways and protein dysfunction (Finding F002). The central lesion is corruption of the IKAROS-family transcription-factor network. Mutant AIOLOS homodimers and AIOLOS–IKAROS heterodimers *"did not bind the canonical AIOLOS-IKAROS DNA sequence"* (PMID: 34155405). The authors define the new principle: *"Heterodimeric interference is a new mechanism of autosomal dominance that causes inborn errors of immunity by impairing protein function via the mutation of its heterodimeric partner"* (PMID: 34155405). Functionally, the mutant acts dominant-negative over WT AIOLOS but not WT IKAROS, yet corrupts IKAROS function by sequestering it into non-functional complexes (PMID: 34694366).

Cellular processes. AIOLOS regulates lymphocyte development, B-cell maturation, plasma-cell generation, and T-helper differentiation. It also targets pericentromeric heterochromatin, linking it to chromatin-mediated gene silencing — a function lost by the DBD mutants.

Immune system involvement. The disease is fundamentally an immunodeficiency with an autoimmune/dysregulation component — reflecting AIOLOS's dual roles in building protective immunity and enforcing tolerance.

T-cell programs (Finding F006). Beyond B cells, AIOLOS shapes T-helper biology: under Th17-polarizing conditions STAT3 and AhR upregulate Aiolos, which *"silenced the Il2 locus, promoting T(H)17 differentiation in vitro and in vivo"* (PMID: 22751139). Aiolos also *"promotes CXCR3 expression on Th1 cells by propagating the IFN-γ/STAT1 cytokine signaling pathway"* (PMID: 39560988) and regulates eosinophil tissue homing via CCR3 (PMID: 34341502).

Drug-target parallel (Finding F005). AIOLOS/IKZF3 is a cereblon neosubstrate. CELMoDs/IMiDs act as molecular glues: *"recruiting novel protein substrates for selective proteasomal degradation. These include two critical lymphoid transcription factors, Ikaros family zinc finger proteins 1 and 3 (IKZF1 and IKZF3), also known as Ikaros and Aiolos"* (PMID: 38539501). Pharmacologic degradation of Aiolos *"correlates with increased transcription of interferon (IFN)-stimulated genes"* (PMID: 26002965) — showing Aiolos normally represses interferon-stimulated genes. This drug-induced degradation is the pharmacologic analog of the genetic deficiency.

Suggested ontology terms. - GO biological process: regulation of lymphocyte differentiation (GO:0045619); B cell differentiation (GO:0030183); T-helper 17 cell differentiation (GO:0072539); negative regulation of transcription, DNA-templated (GO:0045892); regulation of immune response (GO:0050776). - GO cellular component: nucleus (GO:0005634); heterochromatin / pericentric heterochromatin (GO:0005721); transcription regulator complex (GO:0005667). - CL cell types: B cell (CL:0000236); plasma cell (CL:0000786); T-helper 17 cell (CL:0002038); T-helper 1 cell (CL:0000545); naive T cell (CL:0000898); eosinophil (CL:0000771). - CHEBI (drug analogs): lenalidomide (CHEBI:63791); pomalidomide; thalidomide (CHEBI:9513).

---

7. Anatomical Structures Affected

Organ / body-system level. The **immune/hematolymphoid system** is primarily affected (UBERON:0002405 immune system; UBERON:0002390 hematopoietic system). Secondary organ involvement follows infection: **lungs/respiratory tract** (UBERON:0002048 lung; recurrent sinopulmonary infection, *Pneumocystis* pneumonia) and the **upper respiratory tract/sinuses**. **Bone marrow** (UBERON:0002371) is central to B-lymphopoiesis and plasma-cell maintenance.

Tissue / cell level. Affected cell populations: **B cells** (CL:0000236) — abnormal maturation with ↑CD21low, absent CD23; **plasma cells** (CL:0000786) — AIOLOS required for long-lived bone-marrow plasma cells; **T cells** (CL:0000084) with naive skewing; **Th17** (CL:0002038) and **Th1** (CL:0000545) subsets; **eosinophils** (CL:0000771).

Subcellular level. The **nucleus** (GO:0005634) and **pericentromeric heterochromatin** (GO:0005721) — the sites of AIOLOS transcriptional/chromatin function.

Localization / lateralization. Systemic (not lateralized); infections typically bilateral (e.g., pneumonia).

---

8. Temporal Development

Onset. Variable and variant-dependent (Finding F008). Dominant-negative DBD variants tend to present early (pediatric) as combined immunodeficiency; haploinsufficiency variants present later with a CVID-like course. Onset pattern is typically **chronic/insidious**, punctuated by acute infections.

Progression. Chronic, lifelong. Course is characterized by recurrent/episodic infections, potential development of autoimmunity, and an increasing risk of B-cell malignancy over time. No formal staging system exists.

Patterns / critical periods. No spontaneous remission; management is lifelong. The principal window for definitive intervention is early identification enabling immunoglobulin replacement, infection prophylaxis, and consideration of hematopoietic stem-cell transplantation before severe infectious or malignant complications accrue.

---

9. Inheritance and Population

Inheritance. Autosomal dominant (heterozygous variants) (Finding F001). Reported missense DBD variants are **fully penetrant**; the N160S variant *"was fully penetrant"* (PMID: 34694366).

Penetrance / expressivity. High (complete for reported DBD missense variants); expressivity is variable, strongly influenced by variant mechanism (DN vs haploinsufficiency).

Epidemiology. Ultra-rare; only a small number of families/individuals reported worldwide. No reliable prevalence or incidence estimates exist. *(Not available — too few cases.)*

Founder effects / consanguinity / carrier frequency. Not applicable — the disease is dominant with de novo or vertically transmitted private variants; extreme gnomAD LoF intolerance (pLI ≈ 1) means carriers of LoF variants are essentially absent from the general population (Finding F009).

Population demographics. No ethnic predilection, geographic clustering, or sex bias has been established given the small number of cases. *(Not available.)*

---

10. Diagnostics

Clinical/laboratory tests. Immunophenotyping by flow cytometry is central and reveals the characteristic signature: increased **CD21low B cells**, **absent CD23** expression, abnormal B-cell maturation, abrogated CD40 responses, increased naive T cells, and abnormal T-cell differentiation (PMID: 34694366). Serum immunoglobulins typically show impaired antibody responses/hypogammaglobulinemia. Functional assays include pericentromeric heterochromatin localization and electrophoretic mobility shift assays (EMSA) to confirm DNA-binding defects — approaches validated for the IKZF family (PMID: 38734037).

Genetic testing. Definitive diagnosis is molecular. **Whole-genome sequencing (WGS)** or **whole-exome sequencing (WES)** is the recommended approach; the primary atopic/IEI literature supports upfront genomic sequencing for suspected monogenic IEI (PMID: 39381601). A companion IKZF-family report emphasizes that panels may miss variants and that WGS can detect pathogenic IKZF variants where panels and exomes were non-diagnostic (PMID: 38734037). *IKZF3* should be included in IEI/CVID/combined-immunodeficiency gene panels; single-gene testing confirms a known familial variant.

Clinical criteria / differential diagnosis. No disease-specific criteria. Differential diagnosis includes other IKZF-associated IEIs (**IKZF1/IKAROS** — CVID-like or combined immunodeficiency; **IKZF2/HELIOS**), CVID of other genetic cause, and other combined immunodeficiencies. The distinguishing feature is the *IKZF3* variant plus the heterodimeric-interference functional signature.

Screening. Cascade genetic testing of at-risk relatives once a familial variant is identified. No population newborn screening exists for this ultra-rare disorder.

---

11. Outcome / Prognosis

Survival / mortality. No formal survival statistics exist. Prognosis is driven by infection severity (combined immunodeficiency with *Pneumocystis* pneumonia carries significant risk) and malignancy. *(Quantitative survival data not available.)*

Morbidity / complications. Recurrent and opportunistic infections, autoimmunity, and B-cell malignancy — a family was reported *"with T and B cell involvement, Pneumocystis jirovecii pneumonia, and/or chronic lymphocytic leukemia"* (PMID: 34694366). Lifelong immunoglobulin dependence contributes to morbidity.

Prognostic factors. The variant mechanism is the strongest prognostic determinant (Finding F008): **dominant-negative DBD variants** → early-onset combined immunodeficiency, high *Pneumocystis* risk, naive-skewed T cells — worse prognosis; **haploinsufficiency variants** → milder CVID-like course. Across the IKZF family, *"Mutations acting by haploinsufficiency present with a common variable immune deficiency-like phenotype mainly characterized by increased susceptibility to infections. Mutations acting in a dominant negative fashion present with a combined immunodeficiency phenotype with high prevalence of Pneumocystis jirovecii pneumonia"* (PMID: 32654692).

---

12. Treatment

No IMD84-specific disease-modifying therapy exists; management is supportive and extrapolated from combined immunodeficiency / CVID care.

Intervention	Rationale	Suggested NCIT concept
Immunoglobulin replacement (IVIG/SCIG)	Corrects impaired antibody responses / hypogammaglobulinemia	NCIT: Intravenous Immunoglobulin Therapy
Antimicrobial prophylaxis (e.g., anti-*Pneumocystis*: trimethoprim-sulfamethoxazole)	Prevents opportunistic/recurrent infection	NCIT: Antibiotic Therapy / Prophylaxis
Hematopoietic stem-cell transplantation (HSCT)	Potentially curative for severe combined immunodeficiency	NCIT: Hematopoietic Stem Cell Transplantation
Malignancy surveillance / treatment (e.g., for CLL)	Elevated B-cell malignancy risk	NCIT: Cancer Treatment
Supportive care, vaccination caution (live vaccines)	Symptom and infection management	NCIT: Supportive Care

Pharmacogenomic note. Because AIOLOS is a CRBN neosubstrate, IMiDs/CELMoDs (lenalidomide, pomalidomide, mezigdomide) *degrade* AIOLOS and would be expected to worsen the deficiency — relevant if a patient with IMD84 also had a plasma-cell neoplasm; these agents are generally contraindicated with respect to the underlying immune defect (Finding F005). No genotype-guided IMD84-specific pharmacotherapy is established.

Experimental therapeutics. No IMD84-specific clinical trials (no NCT identifiers) are known.

13. Prevention

Being a highly penetrant monogenic disorder, **primary prevention** centers on **genetic counseling** and reproductive options (prenatal testing, preimplantation genetic diagnosis) for families with a known variant. **Secondary prevention** is early molecular diagnosis via genomic sequencing and cascade testing of relatives, enabling pre-emptive immunoglobulin replacement and infection prophylaxis. **Tertiary prevention** targets complications: anti-infective prophylaxis (notably anti-*Pneumocystis*), vaccination strategy adjusted for

immunodeficiency (avoid live vaccines), malignancy surveillance, and timely HSCT in severe cases. No immunization prevents the disease itself; no behavioral or environmental interventions apply.

---

14. Other Species / Natural Disease

Taxonomy / orthologs. The gene is conserved in mouse (*Mus musculus*, NCBI Taxon 10090) as *Ikzf3* (Aiolos). No naturally occurring companion-animal or wildlife disease equivalent to IMD84 has been documented in OMIA. **Comparative biology** is addressed through engineered models (next section). Evolutionary conservation of the AIOLOS/IKAROS dimerization and DNA-binding architecture underlies the faithful recapitulation of human phenotypes in mice. No zoonotic potential (the disease is genetic, not transmissible).

---

15. Model Organisms

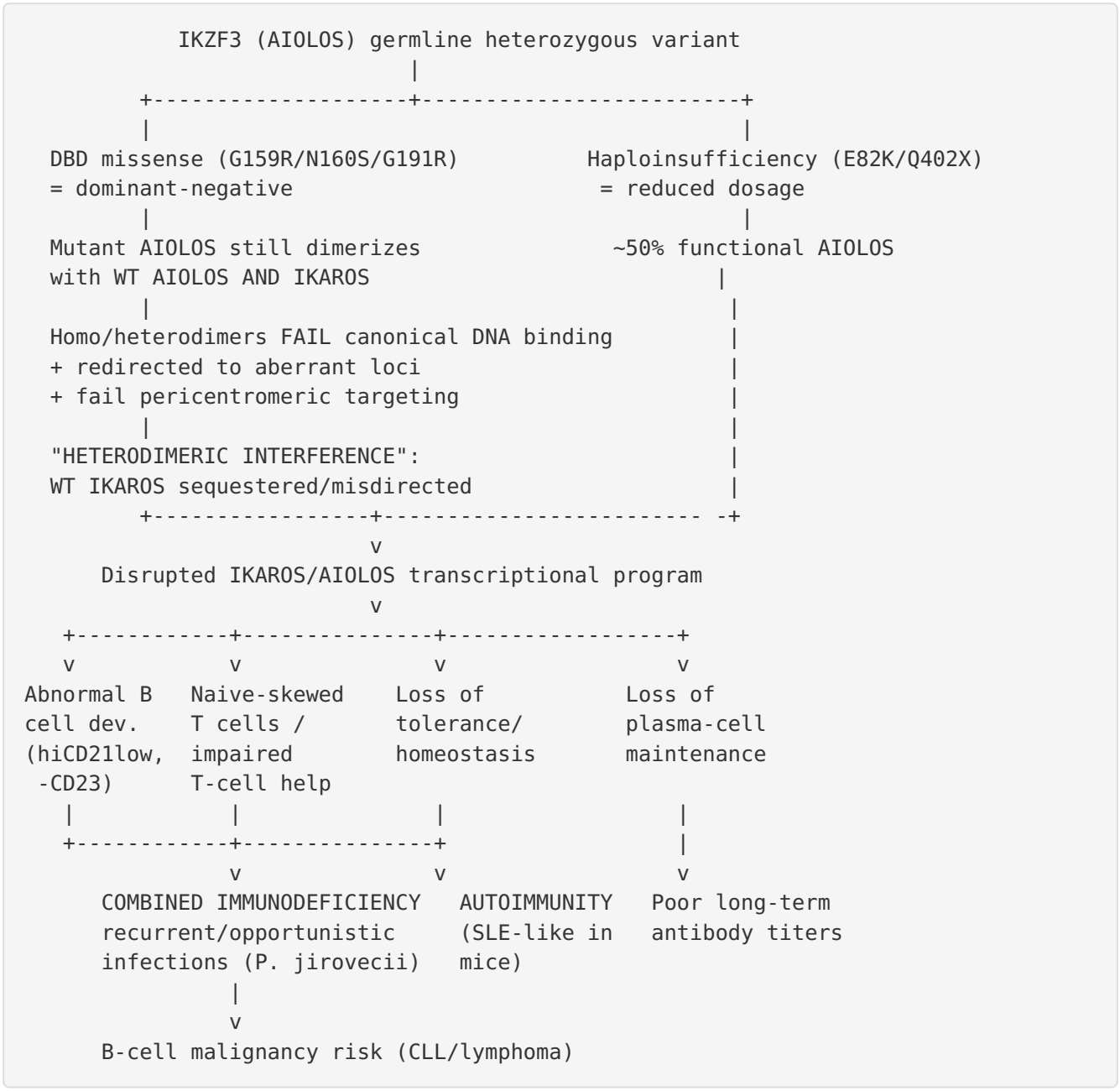
Knock-in models of human variants (Finding F004). Mice carrying the corresponding human N160S (Kuehn) and G159R (Yamashita) variants recapitulate the human B- and T-cell phenotypes: *"The human immunophenotype was recapitulated in a murine model carrying the corresponding human mutation"* (PMID: 34694366). These are the most disease-faithful models.

Classic knockout models. *Ikzf3*-null mice reveal the broader biology: - B cells show an activated phenotype with augmented BCR proliferation, spontaneous germinal centers, elevated IgG/IgE, and autoantibodies; *"Auto-antibodies and development of B cell lymphomas are frequently seen among aging Aiolos mutants"* (PMID: 9806640) — modeling the autoimmunity and malignancy risk. - Aiolos is required for the humoral memory compartment: *"the Ikaros family member, Aiolos, is specifically required for the generation of these plasma cells"* (long-lived high-affinity bone-marrow plasma cells) (PMID: 14718515). - Aiolos-null mice develop SLE-like anti-dsDNA antibodies and immune-complex glomerulonephritis (PMID: 12574333).

Cereblon-humanized models. Studying the AIOLOS–CRBN drug axis requires humanized-CRBN mice because *"a single amino acid substitution of valine to isoleucine in mouse CRBN at position 391"* renders *"mice...not susceptible to IMiD-induced degradation of neosubstrates"* (PMID: 33186626).

Model limitations. Mouse CRBN resistance to IMiDs limits pharmacologic modeling without humanization; knockout mice do not capture the dominant-negative heterodimeric-interference mechanism (which requires the specific knock-in). Resources: MGI, IMPC/IMSR for *Ikzf3* alleles; the DT40 chicken B-cell line for targeted functional studies (PMID: 17623906).

Mechanistic Model / Interpretation



Severity rule (genotype–phenotype): DBD dominant-negative variants → combined immunodeficiency + high *Pneumocystis* risk (worse). Haploinsufficiency → CVID-like (milder). AIOLOS N160 is the structural homolog of IKAROS N159, and the paradigm transfers directly from IKZF1 (PMID: 32654692; PMID: 29889099).

Evidence Base

PMID	Title (abbrev.)	Supports
34155405	*A variant in human AIOLOS impairs adaptive immunity by interfering with IKAROS*	Heterodimeric interference mechanism; loss of canonical DNA binding (F002)
34694366	*T and B cell abnormalities, pneumocystis pneumonia, CLL with AIOLOS defect*	N160S full penetrance, dominant-negative, B/T phenotype, PCP, CLL, mouse recapitulation (F001, F003, F004)
38773004	*AIOLOS-Associated Inborn Errors of Immunity*	Full variant spectrum; sinopulmonary infection phenotype (F001, F003)
32654692	*IKAROS Family Zinc Finger 1-Associated Diseases in PID*	DN vs haploinsufficiency genotype-phenotype paradigm (F008)
29889099	*Dominant-negative IKZF1 mutations cause T, B, myeloid combined immunodeficiency*	Naive T-cell skewing in DN DBD variants (F008)
38539501	*Mezigdomide — CRBN E3 ligase modulator*	AIOLOS/IKZF3 is a CRBN neosubstrate (F005)
33186626	*Lenalidomide-sensitive syngeneic murine model (CRBN humanization)*	Species difference in CRBN; need for humanized models (F005)
26002965	*CC-122 mimics an interferon response (DLBCL)*	Aiolos represses interferon-stimulated genes (F005)
22751139	*Aiolos promotes TH17 differentiation by silencing Il2*	Th17/IL-2 silencing role (F006)
39560988	*Aiolos promotes CXCR3 on Th1 via IFN-γ/STAT1*	Th1 migration program (F006)
14718515	*Aiolos required for high-affinity bone-marrow plasma cells*	Plasma-cell/long-term immunity role (F004)
9806640	*Aiolos regulates B cell activation and maturation*	Autoantibodies, B-cell lymphoma in Aiolos-null mice (F004)
12574333	*OBF-1 and SLE-like phenotype in Aiolos mutant mice*	SLE-like autoimmunity in Aiolos-null mice (F004/F006)
38734037	*Novel IKZF1 variant in AD CVID; WGS*	

PMID	Title (abbrev.)	Supports
		WGS utility; haploinsufficiency functional testing (diagnostics)
39381601	*Rapid identification of primary atopic disorders via WGS*	Upfront genomic sequencing for IEI (diagnostics)
42169761	*IKZF-associated inborn errors of immunity*	IKZF family biology/disease context
34341502	*Aiolos regulates eosinophil migration*	CCR3-dependent eosinophil homing (F006)
17623906	*DT40 mutants: transcriptional regulation of B-cell development*	Functional model system for Aiolos (model organisms)

Evidence source types: human clinical/genetic (patient families — 34694366, 38773004, 38734037); functional in vitro (EMSA, localization — 34155405); model organism (knock-in and knockout mice — 34694366, 9806640, 14718515, 12574333, 33186626); computational/population genetics (gnomAD constraint — F009).

Limitations and Knowledge Gaps

1. **Very small patient numbers.** Only a handful of families/individuals are reported, so prevalence, incidence, sex ratio, age distribution, survival, and quality-of-life metrics are unavailable. All phenotype frequencies are qualitative.
2. **Haploinsufficiency phenotype less characterized.** The E82K and Q402X haploinsufficiency arm rests on fewer cases; its mechanism is partly inferred by analogy to IKZF1 and from extreme gnomAD LoF intolerance rather than fully demonstrated patient-level functional data.
3. **No natural animal disease.** Comparative/veterinary data are absent; all animal evidence is from engineered models.
4. **Treatment evidence is extrapolated.** No IMD84-specific trials exist; management recommendations are inferred from combined immunodeficiency/CVID practice.
5. **Modifiers and epigenetics unresolved.** Modifier genes, epigenetic contributions, and determinants of variable expressivity are not defined.

6. **Malignancy risk not quantified.** CLL/lymphoma is reported but the magnitude of lifetime risk is unknown.

Proposed Follow-up Experiments / Actions

1. **International patient registry & natural-history study** for AIOLOS/IKZF3 variants to quantify penetrance, infection burden, autoimmunity, malignancy incidence, and outcomes stratified by variant mechanism.
2. **Systematic functional classification** of all reported and future *IKZF3* variants (DNA-binding EMSA, pericentromeric localization, dimerization assays) to assign DN vs haploinsufficiency and predict severity prospectively.
3. **Deep immunophenotyping / single-cell multi-omics** (scRNA-seq + ATAC) of patient B and T cells to map the disrupted AIOLOS/IKAROS regulatory network at cell-type resolution and define biomarkers.
4. **HSCT outcome analysis** across reported cases to establish indications, timing, and efficacy for severe combined-immunodeficiency presentations.
5. **Cereblon-humanized knock-in mouse studies** combining the N160S/G159R alleles with humanized CRBN to dissect the intersection of genetic deficiency and IMiD pharmacology — clarifying contraindications and potential therapeutic windows.
6. **Structural studies** (cryo-EM / AlphaFold-guided) of mutant AIOLOS–IKAROS heterodimers on aberrant DNA to understand how DBD mutations redirect genomic targeting.

Report compiled from 9 confirmed findings and 29 reviewed papers across a 5-iteration investigation. Causal gene confirmed: IKZF3 (AIOLOS).
